

# Head-Worn Markerless Augmented Reality Inside A Moving Vehicle

Zhiwei Zhu

Mikhail Sizintsev

Glenn Murray

Han-Pang Chiu\*

Ali Chaudhry

Supun Samarasekera

Rakesh Kumar

SRI International

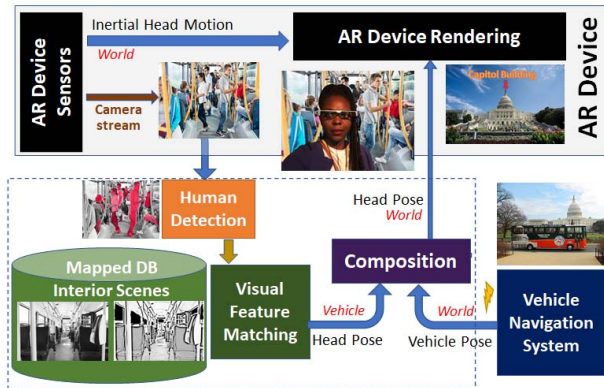


Figure 1: Head-worn outdoor augmented reality (AR) inside a moving vehicle: Provide precise and stable AR of exterior scenes via a head-mounted display (HMD) for a user inside a moving vehicle.

## ABSTRACT

This paper describes a system that provides general head-worn outdoor augmented reality (AR) capability for the user inside a moving vehicle. Our system follows the concept of combining pose estimation from both vehicle navigation system and wearable sensors to address the failure of commercial AR devices inside a moving vehicle. We continuously match natural visual features from the camera against a prebuilt database of interior vehicle scenes. To improve the robustness in a moving vehicle with other passengers, a human detection module is adapted to filter out people from the camera scene. Experiments demonstrate the effectiveness of the proposed solution.

## 1 INTRODUCTION

Augmented reality (AR) can enhance our experience in road traveling and public transportation, by inserting precise synthetic visual information in the exterior environment which is viewed by the user inside the moving vehicle. For example, it can serve as a tour guide to bus passengers by annotating famous buildings in the world.

Providing this AR capability inside a moving vehicle is a challenging problem. Most of commercial wearable AR devices, such as Microsoft HoloLens [2], rely on visual-inertial head tracking that fails under this situation because it creates inconsistency between stationary visual observations and actual user motion [3]. The solution to this problem is to decompose the head pose estimation into two components: the user's head pose in the vehicle is estimated using wearable sensors such as cameras, while a vehicle navigation system computes the vehicle's pose in the real world. However, past systems [5] designed for this solution is limited in their applicability.

\*The contact author is Han-Pang Chiu (han-pang.chiu@sri.com)

These systems assume only one person is in the vehicle, and require the installation of fiducial markers or ranging beacons inside the vehicle beforehand. Also, these marker-based systems cannot continue to provide outdoor augmented reality capability when the user steps out of the vehicle.

In this paper, we propose a new system (Fig. 1) that provides general head-worn outdoor augmented reality (AR) capability for the user inside a moving vehicle. Our system combines pose estimation from both vehicle navigation system (vehicle pose in world) and wearable sensors (head pose in vehicle), and integrates cutting-edge technologies to advance its overall functionalities. In contrast to marker-based systems, we utilize state-of-the-art visual mapping techniques, derived from SLAM algorithms, to build a database of mapped 2D-3D nature visual features from interior vehicle scenes beforehand. This visual database can be reused in other vehicles with similar interior structures. The user can then move freely inside the vehicle, by matching features observed from wearable camera to this database that aligns user's head in a referenced vehicle coordinate system. Furthermore, to avoid outliers in our matching process caused by other passengers in the vehicle, we incorporate human detection technique to identify and segment out people perceived in visual scene. We integrated our system within the Microsoft HoloLens [2], as one of the best available wearable AR devices.

## 2 SYSTEM FRAMEWORK

Figure 1 shows the overview of our proposed system. Our system first decouples the camera and inertial streams from AR devices (HoloLens in our implementation), and then integrates three different kinds of pose/motion measurements to provide precise head pose of the user in the world for wearable AR rendering: (1) Inertial head motion in the world (directly from AR device), (2) Vehicle pose in the world (from vehicle navigation system), (3) Head pose in the vehicle (from human detection and visual feature matching with a database of mapped interior scenes; can also be executed on external server to increase speed and save power consumption on AR device).

### 2.1 Inertial Head Motion in the World

To address the failure of HoloLens inside a moving vehicle, our system turns off (block) all four environment understanding cameras to force HoloLens's visual-inertial fusion module to receive only inertial head motion information (relative to the world). Therefore, the fusion module degrades to IMU-only mode. For orientation estimation, roll and pitch angles are directly observable from the gravity vector, while yaw angle will drift relatively fast. To avoid the drift, constant correction from visual measurements is required from our visual feature matching module.

### 2.2 Vehicle Pose in the World

Our system integrates global vehicle pose  $P_{world \rightarrow veh}$ , which is estimated from a separate vehicle navigation system installed on the vehicle itself, to the AR device. Note vehicle navigation is a well-studied field, and many different kinds of systems have been developed. For our experiments, we relied on GEONYX system [1] – a high-performance inertial navigation system with GNSS extension.

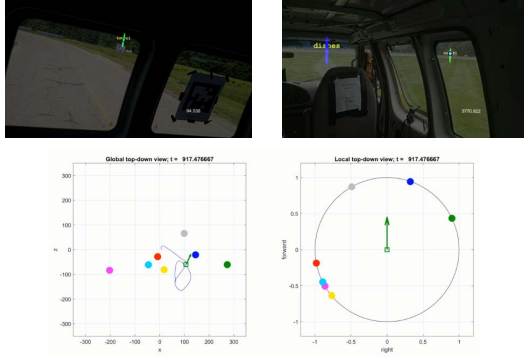


Figure 2: A snapshot of the user's HoloLens display (top two examples), and correspondent estimated head location (bottom-left) and viewpoint (bottom-right) in the world relative to seven referenced landmarks represented with different colors.

### 2.3 Head Pose in the Vehicle

Our system adapts state-of-the-art visual feature matching mechanism [6] to compute precise user head pose relative to the vehicle  $P_{veh \rightarrow head}$ . Visual features perceived from HoloLens frontal HD video camera are extracted and matched against a pre-built database of mapped 2D-3D features from interior vehicle scenes.

The database essentially consists of 2D visual features with descriptors, image locations and 3D coordinates computed from triangulation. It is built offline using Intel RealSense D435i suit that has IMU and stereo camera using method adapted from [6], which has demonstrated its constructed database can be used across variations in lighting as well as different mapped instances of the same model. Therefore, we expect our constructed database can be transferred to other vehicles of same make model with common interior structures.

Considering other passengers inside the same vehicle may hamper the feature matching mechanism, we build a module that segments out people from the images using state-of-the-art deep learning techniques – features detected on human segments are simply excluded from the  $P_{veh \rightarrow head}$  estimation process. Considering the speed requirement, we choose the YOLACT [4] as the base of our human detection module and retrain it to focus only on the human class.

By combining with the vehicle pose  $P_{world \rightarrow vehicle}$  estimated from vehicle navigation system (Sect. 2.2), we compute the actual head pose relative to the world  $P_{world \rightarrow head}$  as follows. For each successful interior match, the computed  $P_{veh \rightarrow head}$  adjusts the global position (by within-vehicle correction) but, more importantly, adjusts orientation that effectively eliminates the drift in yaw. It also puts the HoloLens in the corrected world coordinate system.

$$P_{world \rightarrow head} = P_{veh \rightarrow head} * P_{world \rightarrow veh} \quad (1)$$

When interior match does not occur, we only update the head pose relative to the world  $P_{world \rightarrow head}$  by correcting the global location in  $P_{world \rightarrow veh}$  which is estimated by vehicle navigation system. Any global orientation change in  $P_{world \rightarrow head}$  will be automatically computed by HoloLens high-frequency IMU.

Inside our Unity application, the orientation and translation correction to  $P_{world \rightarrow head}$  are realized as adjustments to the external world coordinate frame rather than the head pose itself – this allows us to unobstructively use HoloLens's real-time IMU-based head movement prediction capability as well as fast synchronized display rendering technique, that guarantees a stable placement of the inserted symbols or objects over the physical world.

### 3 EXPERIMENTAL RESULTS

Experiments were conducted in a van which has back and one side window to verify correctness of outdoor marker insertions. Our head

Table 1: The RMS accuracy improvements with human detection.

| Setting              | Position Error | Orientation Error |
|----------------------|----------------|-------------------|
| No Human Detection   | 13.587         | 3.385             |
| With Human Detection | 12.136         | 2.368             |
| Improvement          | 10.68%         | 30.04%            |

pose estimation software is integrated into the HoloLens device, and the graphics renderer application is developed using Unity and designed to display AR markers with optional text annotations in a global world coordinates. Our application also receives external vehicle pose information (vehicle location and orientation in the world) from the vehicle navigation system via wireless communication.

Seven referenced landmarks, which correspondent to specific buildings or entities in the real world, were selected at a testing site within 500-meter radius. The user can rotate their head freely inside the moving vehicle to observe each AR marker at its physical world locations outside. During the tests, our driving speed is between 20-40 MPH (miles per hour). Figure 2 (top) shows snapshots of the user's view in the HoloLens display. The virtual blue marker "dishes" is aligned well with the real satellite dishes in the world.

We designed several ways to quantitatively evaluate different perspectives of our system to enable AR insertions in the world. The estimated yaw drift from inertial-only HoloLens tracking was more than 12 degrees after running for around 200 seconds. Based on this evaluation, our visual matching process is required to correct the drift from the inertial motion in HoloLens. Since our visual feature matching process is not affected by the movement of the vehicle, we perform this accuracy evaluation inside a stationary vehicle. This allows us to compare the recovered  $P_{vehicle \rightarrow head}$  against a pseudo ground-truth generated using the HoloLens (with full visual-inertial fusion mode, since the vehicle is stationary). Table 1 shows the computed RMS accuracy comparison for feature matching module with and without human detection capability being used. From the experimental results, the use of human detection is able to improve 30% RMS accuracy in yaw orientation estimation, that is key for accurate AR insertion in the exterior world.

To evaluate the performance of our entire system, the user's HoloLens viewing was recorded when the user sits inside the moving vehicle and looks around to observe the "tower" through the windows as much as possible during the live tests. For evaluation, we manually selected those frames that the real "tower" was observed through the window from the recorded video. We then extracted the locations of both the real "tower" and the green virtual "tower" AR insertions in the images manually (as depicted in Fig. 2) and measured their pixel discrepancy. Since camera calibration is known, we convert the computed pixel deviations into angular errors. There is total of 165 frames when the user was able to observe the real "tower" through the van windows. The computed mean and root-mean-square error (RMSE) is  $0.8407^\circ$  and  $1.1296^\circ$  respectively, and the 90-percentage error is  $1.8840^\circ$ .

### ACKNOWLEDGEMENTS

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